Exercise 13

On the definite integral

From the school book | Understand | Apply | Higher Order Thinking Skills

1 From the school book: Find the definite integration to the function f from $x = -2$ to $x = 4$ where $f(x) = 3x^2 - 2$ « 60 »

2 Find the value of each of the following :

(1) $\int_1^4 7 \, dx$ « 21 »

(2) From the school book: $\int_{-1}^2 (2x + 3) \, dx$ « 12 »

(3) $\int_{-1}^3 (4x^3 - 6x^2 + 5) \, dx$ « 44 »

(4) $\int_1^3 \frac{6 - x^4}{2x^2} \, dx$ « $\frac{-7}{3}$ »

(5) $\int_0^5 \sqrt{3x + 1} \, dx$ « 14 »

(6) From the school book: $\int_0^4 (2x + 1)^{\frac{3}{2}} \, dx$ « $\frac{242}{5}$ »

(7) From the school book: $\int_0^5 \frac{3}{\sqrt{t + 4}} \, dt$ « 6 »

(8) From the school book: $\int_4^0 \frac{dt}{\sqrt{8 - t}}$ « $4 - 4\sqrt{2}$ »

(9) $\int_1^4 \frac{x - 1}{\sqrt{x} - 1} \, dx$ « $\frac{23}{3}$ »

(10) $\int_{-1}^1 \frac{x + 8}{\sqrt[3]{x} + 2} \, dx$ « $\frac{46}{5}$ »

3 Find the value of each of the following :

(1) $\int_0^{\pi} \sin x \, dx$ « 2 »

(2) From the school book: $\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \cos \theta \, d\theta$ « 2 »

(3) $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \csc \theta \cot \theta \, d\theta$ « Zero »

(4) $\int_0^{\frac{\pi}{3}} \sec^2 x \, dx$ « $\sqrt{3}$ »

(5) From the school book: $\int_0^1 2 \sin \pi z \, dz$ « $\frac{4}{\pi}$ »

(6) From the school book: $\int_{-\pi}^{\pi} (4 + \pi \cos 2x) \, dx$ « 8π »

(7) $\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (\sin x + \cos x) \, dx$ « 2 »

(8) From the school book: $\int_0^{\frac{\pi}{4}} \tan z \sec^2 z \, dz$ « $\frac{1}{2}$ »

(9) $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{dx}{\sin^2 x}$ « 1 »

(10) $\int_0^{\frac{\pi}{3}} \frac{\sin x}{\cos^2 x} \, dx$ « 1 »

4 Find the value of each of the following :

(1) $\int_1^e \frac{2}{x} \, dx$ « 2 »

(2) $\int_e^1 \frac{dx}{x}$ « – 1 »

(3) $\int_{-5}^{-1} \frac{3}{x} \, dx$ « $-3 \ln 5$ »

(4) $\int_0^{\ln 3} e^x \, dx$ « 2 »

(5) $\int_0^3 (2x - 7e^x)\,dx$ « $16 - 7e^3$ »

(6) $\int_{\ln 2}^{\ln 4} (2e^{2x} - e^x)\,dx$ « 10 »

5 **Express each of the following using one integration :**

(1) $\int_0^5 x^2\,dx + \int_{-2}^0 x^2\,dx$

(2) $\int_{-4}^{-1} \sqrt{x+6}\,dx - \int_4^{-1} \sqrt{x+6}\,dx$

(3) $\int_2^6 k\,dx + \int_3^2 k\,dx$

(4) $\int_0^4 2x^2\,dx + \int_4^5 x^2\,dx$

6 If $\int_2^4 f(x)\,dx = 5$, $\int_2^4 g(x)\,dx = -3$, **then find :**

(1) $\int_2^4 3f(x)\,dx$ (2) $\int_4^2 2g(x)\,dx$

(3) $\int_2^4 [3f(x) - 2g(x)]\,dx$ (4) $\int_2^4 [3g(x) + 4x]\,dx$ « 15 , 6 , 21 , 15 »

7 If $\int_{-2}^3 f(x)\,dx = 9$, $\int_5^3 f(x)\,dx = 4$, **then find the value of :**

$\int_{-2}^5 [3f(x) - 6x]\,dx$ « – 48 »

8 If the function f is continuous on $\mathbb{R}$ and $\int_{-1}^4 f(x)\,dx = 255$, $\int_2^{-1} f(x)\,dx = -15$

Find : $\int_2^4 f(x)\,dx$ « 240 »

9 If $f(x) = \begin{cases} x^2 & , \ x < 2 \\ 3x - 2 & , \ x \geq 2 \end{cases}$ **Find :** $\int_0^3 f(x)\,dx$ « $\frac{49}{6}$ »

10 If $f(x) = \begin{cases} 2 & , \ x < 2 \\ x & , \ x \geq 2 \end{cases}$ **Find :** $\int_0^6 f(x)\,dx$ « 20 »

11 *(2nd Session 2019)* If $f(x) = \begin{cases} 2x + x^2 & , \ x < 0 \\ 2x - x^2 & , \ x \geq 0 \end{cases}$ **Find :** $\int_{-1}^3 f(x)\,dx$ « $\frac{-2}{3}$ »

12 Find the value of each of the following :

(1) $\int_{-1}^{3} |x-1| \, dx$ « 4 »

(2) $\int_{0}^{5} |x-3| \, dx$ « $\frac{13}{2}$ »

(3) $\int_{0}^{2} |2x-1| \, dx$ « $\frac{5}{2}$ »

(4) $\int_{-3}^{1} \sqrt{x^2+4x+4} \, dx$ « 5 »

(5) $\int_{-1}^{3} x|x| \, dx$ « $8\frac{2}{3}$ »

(6) $\int_{-1}^{3} 3x|x-4| \, dx$ « 20 »

(7) $\int_{-3}^{3} |x^2-4| \, dx$ « $15\frac{1}{3}$ »

(8) $\int_{-1}^{5} |x^2-4x| \, dx$ « $15\frac{1}{3}$ »

13 Choose the correct answer from the given ones :

(1) If $\int_{1}^{3} f(x) \, dx = 5$, $\int_{3}^{1} g(x) \, dx = 7$

, then $\int_{1}^{3} (4f(x) + g(x) + 3) \, dx =$ .....

(a) – 12 (b) – 7 (c) 12 (d) 19

(2) $\int_{a}^{b} f(x) \, dx =$ .....

(a) $f(b) - f(a)$ (b) $\int_{b}^{a} f(x) \, dx$

(c) $-\int_{b}^{a} f(x) \, dx$ (d) b – a

(3) If $\int_{2}^{5} f(x) \, dx = 4$, then $\int_{2}^{5} [3f(x) - 1] \, dx =$ .....

(a) 9 (b) 11 (c) 12 (d) – 8

(4) If $\int_{-2}^{3} f(x) \, dx = 12$, $\int_{-2}^{5} f(x) \, dx = 16$, then $\int_{3}^{5} f(x) \, dx =$ .....

(a) – 28 (b) – 4 (c) 4 (d) 28

(5) If the function f is continuous on $\mathbb{R}$, $\int_{-1}^{3} f(x) \, dx = 7$, $\int_{5}^{3} f(x) \, dx = -11$

, then $\int_{-1}^{5} f(x) \, dx =$ .....

(a) – 4 (b) 18 (c) – 18 (d) 77

(6) *(1st session 2017)* If $f(x) = |x|$, then $\int_{-2}^{2} f(x) \, dx =$ .....

(a) –1 (b) zero (c) 2 (d) 4

(7) $\int_{0}^{2} (2 - |x|) \, dx =$ .....

(a) 4 (b) 2 (c) 1 (d) zero

(8) If f' is continuous on the interval $[a , b]$, then $f(a) + \int_a^b f'(x)\,dx =$ .....

(a) $f'(b)$ (b) $f(b)$ (c) $f(a)$ (d) $f'(a)$

(9) $\frac{d}{dx}\left[\int_2^3 x^2\sqrt{x^2+1}\,dx\right] =$ .....

(a) -1 (b) zero (c) 1 (d) 2

(10) ***(2nd Session 2017)*** If $f(x) = \sin^3 x$, then $\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} f(x)\,dx =$ .....

(a) 4 (b) 2 (c) zero (d) -1

(11) $\int_a^b x^2\,dx + \int_b^a y^2\,dy =$ ..... (where $a \neq b$)

(a) $2\int_a^b x^2\,dx$ (b) zero (c) $2\int_b^a y^2\,dx$ (d) $b - a$

(12) If $\int_{-2}^2 f(x)\,dx =$ zero , then $f(x)$ could be .....

(a) $x^2 + 1$ (b) x (c) $x + 1$ (d) $x - 1$

(13) If $\int_{-2}^3 f(x)\,dx + \int_3^7 f(x)\,dx - \int_5^7 f(x)\,dx = \int_a^b f(x)\,dx$, then $a + b$ can be equal to .....

(a) 5 (b) -3 (c) 3 (d) -5

(14) $\int_e^{e^2} \frac{dx}{x} =$ .....

(a) 1 (b) 2 (c) e (d) e^2

(15) If $\int_k^3 2x\,dx = 5$, then $k =$ .....

(a) 13 (b) 5 (c) 1 (d) ± 2

(16) $\int_0^a |x|\,dx = 32$, then $a =$ .....

(a) 8 (b) -8 (c) ± 8 (d) zero

(17) ***(1st Session 2023)*** If $\int_{-k}^k \sqrt{k + x}\,dx = \frac{2}{3}$, then $k =$ .....

(a) 2 (b) $-\frac{1}{2}$ (c) $\frac{1}{2}$ (d) -1

(18) The value of $\int_{-2}^2 (ax^3 + bx + c)\,dx$ depends on .....

(a) value of b (b) value of c (c) value of a (d) value of a , b

(19) If $f(x) = \frac{3x^{2021} + 1}{x^{2023} + 1}$, then $\int_0^1 \grave{f}(x)\,dx =$ .....

(a) zero (b) 1 (c) 2 (d) 3

(20) If f is a differentiable function on $\mathbb{R}$, then $f(4) =$ ..... $+ \int_1^4 f'(x)\,dx$

(a) $f(0)$ (b) $f(1)$ (c) $f'(1)$ (d) $f'(4)$

(21) If $\int_0^{\frac{\pi}{4}} \frac{4+\cos x}{1+x^2}\,dx = k$, then $\int_{\frac{-\pi}{4}}^{\frac{\pi}{4}} \frac{16+4\cos x}{1+x^2}\,dx =$ .....

(a) 2 k (b) 4 k (c) 6 k (d) 8 k

(22) If $\int_1^k x^{n-1}\,dx = \frac{2}{n}$, then k = ...

(a) 3 n (b) 3^n (c) $\frac{3}{n}$ (d) $\sqrt[n]{3}$

(23) If $\int_{-1}^{a} f(z)\,dz = 4a^3 - 2a + c$, then c = ...

(a) – 1 (b) 1 (c) 2 (d) – 2

(24) *(2nd Session 2024)* If the function f is an odd function and continuous on the interval $[-6, 6]$, $\int_4^6 f(x)\,dx = 7$, then $\int_{-6}^{4} [f(x)+x]\,dx =$ .....

(a) – 17 (b) – 3 (c) 3 (d) 17

(25) *(1st Session 2022)* If f is a continuous function on $\mathbb{R}$ and $\int_{-1}^{7} [f(x)-2x]\,dx = 22$, then $\int_{-1}^{7} [2f(x)+x]\,dx =$ .....

(a) 116 (b) 164 (c) 76 (d) 66

---

14 **Use the properties of the odd and even functions to find the value of each of the following :**

(1) $\int_{-3}^{3} \frac{x}{1+x^2}\,dx$ « zero »

(2) $\int_{-2}^{2} \frac{x^3-3x}{x^2+1}\,dx$ « zero »

(3) $\int_{-3}^{3} (x^2-1)\,dx$ « 12 »

(4) $\int_{-2}^{2} \left(x^3\sqrt{x^2+3}\right)\,dx$ « zero »

(5) $\int_{-4}^{4} (4-|x|)\,dx$ « 16 »

(6) $\int_{-2}^{2} x^2(3-2|x|)\,dx$ « zero »

(7) $\int_{-1}^{1} \sqrt{x^4+x^2}\,dx$ « $\frac{4\sqrt{2}-2}{3}$ »

---

15 If $\int_0^2 x^2\,dx = \frac{8}{3}$, **then find the value of each of the following , giving the reason :**

(1) $\int_{-2}^{0} x^2\,dx$ **(2)** $\int_{-2}^{2} x^2\,dx$ **(3)** $\int_0^2 -x^2\,dx$

« $\frac{8}{3}$, $\frac{16}{3}$, $\frac{-8}{3}$ »

---

16 If $\int_0^2 x^3\,dx = 4$, **then find the value of each of the following , giving the reason :**

(1) $\int_{-2}^{0} x^3\,dx$ **(2)** $\int_{-2}^{2} x^3\,dx$ **(3)** $\int_0^2 3x^3\,dx$

« – 4 , 0 , 12 »

17 If function f is continuous on the interval $[-4, 4]$ and $\int_0^4 f(x)\,dx = 3$, then find the value of each of :

(1) $\int_0^4 [f(x) - 2]\,dx$ (2) $\int_{-4}^4 f(x)\,dx$, f odd (3) $\int_{-4}^4 f(x)\,dx$, f even.

« -5 , 0 , 6 »

18 What is the value of $\int_{-3}^5 f(x)\,dx$ if the function f is continuous and odd on the interval $[-3, 5]$, $\int_3^5 f(x)\,dx = 9$? « 9 »

19 What is the value of $\int_{-4}^2 f(x)\,dx$ if the function f is continuous and even on the interval $[-4, 4]$, $\int_{-4}^4 f(x)\,dx = 20$ and $\int_0^2 f(x)\,dx = 6$? « 16 »

20 **Find the value of each of the following :**

(1) $\int_0^2 x(x^2 - 3)^2\,dx$ « $\frac{14}{3}$ »

(2) $\int_0^3 x\sqrt{25 - x^2}\,dx$ « $\frac{61}{3}$ »

(3) $\int_0^4 \frac{x}{\sqrt{9 + x^2}}\,dx$ « 2 »

(4) $\int_0^2 \frac{x^2 + 2x}{\sqrt{x^3 + 3x^2 + 16}}\,dx$ « $\frac{4}{3}$ »

(5) $\int_{-4}^0 x(x + 4)^3\,dx$ « $\frac{-256}{5}$ »

(6) $\int_0^8 \frac{x}{\sqrt{x + 1}}\,dx$ « $\frac{40}{3}$ »

(7) $\int_{-2}^0 y\sqrt[3]{1 + y}\,dy$ « $\frac{6}{7}$ »

(8) $\int_1^e (\ln x)\,dx$ « 1 »

21 **Choose the correct answer from the given ones :**

(1) $\int_0^4 4^{\log_2 x}\,dx =$ .....

(a) $\frac{64}{3}$ (b) $\frac{16}{3}$ (c) 4 (d) $\frac{11}{3}$

(2) If $\int_k^{k+1} (9)^{\log_3 \sqrt{x}}\,dx = \frac{5}{2}$, then k = ...

(a) zero (b) 1 (c) 2 (d) 3

(3) $\frac{1}{\pi}\int_{\frac{2}{\pi}}^{\pi} \frac{\sin\left(\frac{1}{x}\right)}{x^2}\,dx =$ .....

(a) -2 (b) -1 (c) zero (d) 1

(4) $\int_e^{e^3} \frac{(\ln x)^3}{x}\,dx =$ .....

(a) 20 (b) 16 (c) 12 (d) 10

(5) If $\int_{-2}^2 (x^7 + k)\,dx = 16$, then k = ...

(a) -12 (b) -4 (c) zero (d) 4

(6) If $f(x)$ and $\grave{f}(x)$ are continuous functions and $f(6) = 4$, $\grave{f}(6) = 3$, $f(7) = 14$, $\grave{f}(7) = 5$, then $\int_6^7 \grave{f}(x)\, dx =$ .....

(a) 10 (b) 2 (c) 18 (d) 8

(7) *(2nd session 2022)* $\int_0^{2e} \frac{dx}{x+e} =$ .....

(a) $\ln 2e$ (b) $\ln 3$ (c) e (d) 1

(8) If the function f is continuous , $f(5) = 9$, $f(1) = 4$, then $\int_1^5 3\sqrt{f(x)}\ .\ \grave{f}(x)\, dx =$ .....

(a) 19 (b) 38 (c) $10\sqrt{3} - 2$ (d) 50

(9) $\int_0^{\pi} x^2 \cos x\, dx =$ .....

(a) $-\pi$ (b) -2π (c) π (d) 2π

(10) If $\int_0^{\theta} \frac{\sec^2 x}{1+\tan x}\, dx = \ln 2$ where , $0 < \theta < \frac{\pi}{2}$ then $\theta =$ .....

(a) $\frac{\pi}{12}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$

(11) If $\int_1^4 f(x)\, dx + \int_{2b}^8 f(x)\, dx = \int_1^8 f(x)\, dx$, then b can be equal .....

(a) 1 (b) 2 (c) 4 (d) 8

(12) If $\int_0^k (3x^2 - 1)\, dx = k^3 - 2$, then $k =$ .....

(a) 6 (b) 4 (c) 2 (d) 1

(13) If $\int_2^a (4x - 1)\, dx = 9$, $a \in \mathbb{Z}^+$, then $a =$ .....

(a) -3 (b) 1 (c) 3 (d) 4

(14) If $\left(\int_0^a x\, dx\right)^3 = \int_0^a x^3\, dx$, $a \in \mathbb{R}^+$, then $a =$ .....

(a) 2 (b) $\sqrt{2}$ (c) 3 (d) $\sqrt{3}$

(15) If $\int_2^{10} f(x)\, dx + \int_8^{11} f(x)\, dx + \int_{10}^8 f(x)\, dx = 9$, then $\int_2^{11} f(x)\, dx =$ .....

(a) 4.5 (b) 9 (c) 12 (d) 18

(16) $\int_{2019}^{2021} (x - 2020)^2\, dx =$ .....

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) 1 (d) $\frac{4}{3}$

(17) If $\int_3^9 f(x)\,dx = 20$, $\int_5^{11} f(x)\,dx = 25$

, then $\int_3^5 f(x)\,dx - \int_9^{11} f(x)\,dx =$ .....

(a) – 10 (b) – 5 (c) zero (d) 5

(18) If $\int_a^b (3x^2 + 2)\,dx = 96$, $\int_a^b dx = 4$, then a × b = ...

(a) 1 (b) 1.5 (c) 2 (d) 2.5

(19) If $\int_{\ln b}^{\ln a} e^x\,dx = 2$, $a^2 - b^2 = 12$, then a = ...

(a) 4 (b) 6 (c) 8 (d) 12

(20) $\int_0^{\frac{\pi}{6}} \sin^4 x\,dx + \int_{\frac{\pi}{6}}^0 \cos^4 x\,dx =$ .....

(a) $\frac{\sqrt{3}}{4}$ (b) $\frac{-\sqrt{3}}{4}$ (c) $\frac{1}{4}$ (d) $-\frac{1}{4}$

(21) If $a = \frac{\pi}{4}\int_{\frac{\pi}{4}}^{3\frac{\pi}{4}} \csc^2 x\,dx$, $b = \frac{3\pi}{4}\int_{\frac{\pi}{4}}^{\frac{\pi}{4}} \cot^2 x\,dx$, then a + b = ...

(a) – 1 (b) 1 (c) $\frac{\pi}{2}$ (d) $\frac{-\pi}{2}$

(22) *(Trial 2021)* If $a, b \in]0, \frac{\pi}{2}[$ then $\int_a^b \tan^2 x\,dx + \int_b^a \sec^2 x\,dx =$ .....

(a) a – b (b) b – a (c) 1 (d) tan b – tan a

(23) If $f(x)$ is a continuous function in $[-7, 7]$ and $f(-x) = f(x)$

and $\int_{-5}^0 f(x)\,dx = 8$, $\int_5^7 f(x)\,dx = 4$, then $\int_{-5}^7 f(x)\,dx =$ .....

(a) 22 (b) 20 (c) 19 (d) 17

(24) If f is defined on $\mathbb{R}$ and $f(-x) = f(x)$ and $\int_{-5}^1 f(x)\,dx = 24$, $\int_1^5 f(x)\,dx = 16$

, then $\int_0^1 f(x)\,dx =$ .....

(a) – 8 (b) 4 (c) 8 (d) 20

(25) If $f(x)$ is an odd function and continuous on $\mathbb{R}$ and $\int_{-6}^4 f(x)\,dx = -3$

, $\int_0^6 f(x)\,dx = 11$, then $\int_0^4 f(x)\,dx =$ .....

(a) 5 (b) 6 (c) 8 (d) 10

(26) *(2nd Session 2021)* If f is an even function and continuous on $\mathbb{R}$, $\int_0^3 f(x)\,dx = 5$, then $\int_{-3}^3 [f'(x) - f(x)]\,dx =$ .....

(a) 15 (b) – 15 (c) – 10 (d) 10

(27) *(1st Session 2022)* If f is an even continuous function on the interval $[-3, 3]$, $\int_{-3}^3 f(x)\,dx = 20$, $\int_0^2 f(x)\,dx = 8$, then $\int_2^3 f(x)\,dx =$ .....

(a) – 2 (b) 2 (c) 12 (d) 28

(28) *(2nd Session 2022)* If the function f is an even continuous function on the interval $[-k, k]$ and $\int_0^k f(x)\,dx = 2$, then $\int_{-k}^k [f(x) - 2x]\,dx =$ .....

(a) – 4 (b) $4 - 4k^2$ (c) 4 (d) $-4 + 4k^2$

(29) $\int_0^r \pi x\,dx =$ .....

(a) circumference of a circle with radius length r.

(b) half the volume of a sphere with radius length r.

(c) half the circumference of a circle with radius length r.

(d) half the area of a circle with radius length r.

(30) $\int_0^h \pi r^2\,dx =$ .....

(a) volume of a right circular cylinder its height (h) and the radius length of its base (r)

(b) area of a sphere its radius length (h)

(c) lateral area of a right circular cylinder its height (h) and radius length of its base (r)

(d) $\frac{1}{3}\pi h^3 + c$

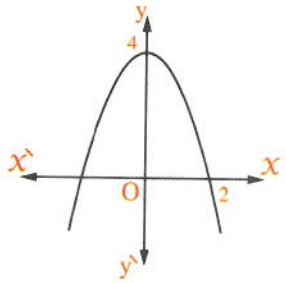
(31) In the opposite figure :

The curve of the function $y = f(x)$

, then $\int_0^2 f'(x)\,dx =$ .....

(a) 2 (b) – 2

(c) – 4 (d) 4

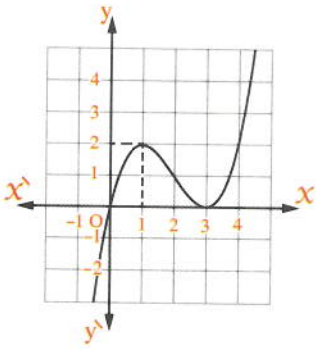


(32) The opposite figure represents the curve of the function $y = f(x)$ and $g(x) = x \cdot f(x)$

, then $\int_1^3 g''(x)\,dx =$ .....

(a) – 1 (b) – 2

(c) – 3 (d) – 4



Higher skills

22 **Choose the correct answer from those given :**

(1) If f is a function where $\int_{6}^{12} f(2x)\,dx = 10$, which of the following is correct .....

(a) $\int_{12}^{24} f(t)\,dt = 5$ (b) $\int_{12}^{24} f(t)\,dt = 20$

(c) $\int_{6}^{12} f(t)\,dt = 5$ (d) $\int_{6}^{12} f(t)\,dt = 20$

(2) If $g(9) = 7$, $g(4) = 3$, then $\int_{2}^{3} 2x\, g(x^2) \cdot \grave{g}(x^2)\,dx =$ .....

(a) 12 (b) 16 (c) 20 (d) 21

(3) $\int_{0}^{20\pi} |\sin x|\,dx =$ .....

(a) 20 (b) 20π (c) 40 (d) 40π

(4) If $\int_{\sin\theta}^{\cos\theta} \frac{x}{2}\,dx = \frac{-\sqrt{3}}{8}$, then one of the values of θ equals .....

(a) 30° (b) 45° (c) 60° (d) 75°

(5) $\int_{1}^{e} e^{x}\left(\frac{1 + x \ln x}{x}\right)dx =$ .....

(a) e (b) e – 1 (c) e^e (d) $e^2 + 1$

(6) If f is a continuous function on $\mathbb{R}$ and $\int_{1}^{5} f(x)\,dx = 7$ and the curve of the function passes through two points $(1, 3)$, $(5, 9)$, then $\int_{1}^{5} x \cdot \grave{f}(x)\,dx =$ .....

(a) 70 (b) 42 (c) 35 (d) 7

(7) If $f(x) = x \ln x$, then $\int_{1}^{e} \grave{f}(x)\,dx =$ .....

(a) 1 (b) 2 (c) $\frac{1}{2}(e^2 - e - 1)$ (d) $\frac{1}{e}(1 - e)$

(8) $\int_{2}^{5} \frac{1}{x+2}\,dx =$ .....

(a) $\int_{0}^{3} \frac{1}{x+4}\,dx$ (b) $\int_{-1}^{2} \frac{1}{x+5}\,dx$

(c) $\int_{3}^{6} \frac{1}{x+1}\,dx$ (d) All the previous.

(9) If $a < 2 < b$ and $\int_{a}^{b} |x - 2|\,dx = 4$, then $\frac{a^2 + b^2}{a + b} =$ .....

(a) $\frac{1}{2}$ (b) 2 (c) 4 (d) – 4

(10) $\int_{0}^{2} (x+6)^2 e^{x^2}\,dx - \int_{0}^{2} (x-6)^2 e^{x^2}\,dx =$ .....

(a) $12\,e^2$ (b) $12\,e^4$ (c) $12(e^4 - 1)$ (d) $24(e^4 - 1)$

(11) If $\int_0^5 f(3x+2)\,dx = 45$, then $\int_0^3 f(5x+2)\,dx =$ .....

(a) 9 (b) 18 (c) 27 (d) 36

(12) If f is a differentiable function on $\mathbb{R}$ and $f(1) = 4$, $f(3) = 10$, then $\int_1^3 [f(x) + x f'(x)]\,dx =$ .....

(a) 22 (b) 24 (c) 26 (d) 28

(13) $\lim_{h \to 0} \left[\frac{1}{h} \int_3^{3+h} \sqrt{x^2+16}\,dx\right] =$ .....

(a) 3 (b) 5 (c) 9 (d) 16

(14) If $\int_0^a (3x^2 - 2x)\,dx \leq 2a$ where $a \in \mathbb{R}^+$, then $a \in$ .....

(a) $[0 , 2]$ (b) $]0 , 2]$ (c) $]2 , \infty[$ (d) $]1 , \infty[$

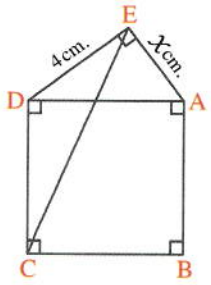
(15) If f is a differentiable twice function in the interval $[a , b]$ and the slope of the tangent of the function f at $x = a$ equals 1 and the slope of the tangent of the function f at $x = b$ equals $\sqrt{2}$, then $\int_b^a 2 f'(x) f''(x)\,dx =$ .....

(a) -2 (b) -1 (c) 1 (d) 2

(16) $\int_0^{10} [(x-1)(x-2)(x-3)\cdots(x-8)(x-9)]\,dx =$ .....

(a) zero (b) 10 (c) 50 (d) 100

(17) In the opposite figure :

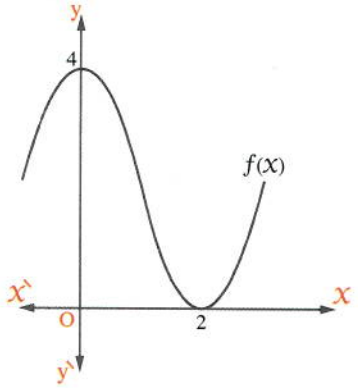


If ABCD is a square , m $(\angle AED) = 90°$, then $\int_0^1 (EC)^2\,dx =$ .....

(a) $\frac{82}{3}$ (b) $\frac{91}{3}$

(c) $\frac{100}{3}$ (d) $\frac{109}{3}$

(18) In the opposite figure :



$\int_0^2 [f(x)]^2 f'(x)\,dx =$ .....

(a) $\frac{-64}{3}$ (b) $\frac{64}{3}$

(c) $\frac{8}{3}$ (d) -64

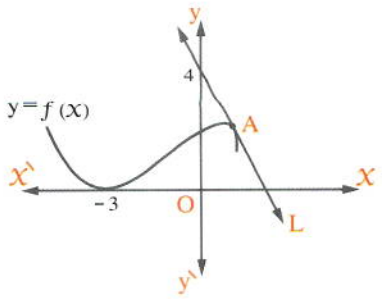
(19) In the opposite figure :

The straight line L touches the curve $y = f(x)$ at the point A $(1 , 2)$

, then $\int_{-3}^{1} f''(x)\, dx =$ .....

(a) -3 (b) -2

(c) -1 (d) 1

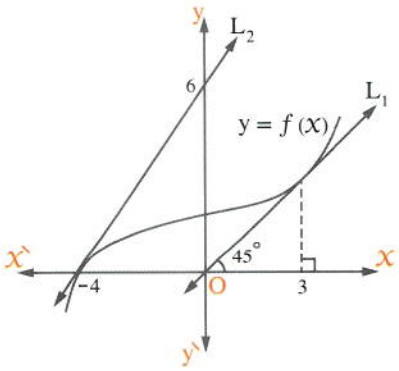


(20) In the opposite figure :

$\int_{-4}^{3} \frac{f''(x)}{f'(x)}\, dx =$ .....

(a) 1 (b) $\ln \frac{3}{2}$

(c) $\ln \frac{2}{3}$ (d) $\log_3 2$



(21) In the opposite figure :

If L touches the curve $y = f(x)$ at the point A $(3 , 3)$

, then $\int_{0}^{3} x\ f''(x)\, dx =$ .....

(a) -2 (b) 2

(c) 4 (d) 8a

